03 Small intestine

Team: _____

Basic function (this is all you need): Most chemical digestion finishes here and most nutrients are absorbed here.

STRUCTURES TO DRAW AND LABEL

- Duodenum: about 25 cm, C shaped around the head of the pancreas, mostly retroperitoneal; superior, descending, inferior (horizontal) and ascending parts
- Major duodenal papilla (bile duct and main pancreatic duct enter at the hepatopancreatic ampulla) and minor duodenal papilla (accessory pancreatic duct)
- Duodenojejunal flexure held by the suspensory muscle of the duodenum (ligament of Treitz)
- Jejunum: proximal two fifths of the mesenteric small intestine, thicker wall, more circular folds
- Ileum: distal three fifths, thinner wall, Peyer's patches, ends at the ileocecal valve
- Mesentery suspends the jejunum and ileum from the posterior abdominal wall
- Surface area features: circular folds (plicae circulares), villi, microvilli (brush border)

NERVES AND BLOOD SUPPLY

- Parasympathetic: vagus nerve (CN X)
- Sympathetic: greater and lesser splanchnic nerves to the celiac and superior mesenteric ganglia
- Enteric plexuses: submucosal (in the submucosa) and myenteric (between the circular and longitudinal muscle). Present along the whole tube.
- Artery: Superior mesenteric artery (jejunal and ileal arteries). The duodenum also gets pancreaticoduodenal branches from the gastroduodenal artery

THE WALL UNDER THE MICROSCOPE

- Simple columnar epithelium with a brush border and goblet cells
- Villi: finger shaped projections with a core of lamina propria, a capillary network and a central lacteal
- Intestinal glands (crypts of Lieberkuhn) open between the villi
- Duodenal (Brunner) glands in the submucosa: found only in the duodenum
- Peyer's patches (aggregated lymphoid nodules) in the ileum
- Muscularis externa: inner circular, outer longitudinal
- Serosa on the jejunum and ileum; adventitia on the retroperitoneal parts of the duodenum

CELLS AND WHAT THEY MAKE

Enterocytes (absorptive cells)	Brush border enzymes on the microvilli; absorb nutrients
Goblet cells	Mucus; numbers increase toward the ileum
Paneth cells (base of the crypts)	Lysozyme and defensins
Enteroendocrine cells	Cholecystokinin (CCK), secretin, and others
Duodenal gland cells	Alkaline mucus
Stem cells (crypts)	Replace the epithelium about every 3 to 6 days
M cells over Peyer's patches	Sample material and pass it to immune cells

TWO MEMORY TOOLS (USE THESE OR MAKE YOUR OWN)

Mnemonic: Folds, Fingers, Fuzz

Circular folds, villi, microvilli, from largest to smallest. Each one multiplies surface area again.

Mnemonic: Brunner's in the Beginning, Peyer's at the Pole end

Duodenal (Brunner) glands mark the beginning of the small intestine; Peyer's patches mark the ileum at the far end. Paneth protects the pit: Paneth cells sit at the bottom of the crypt releasing lysozyme and defensins.